HT410

±0.5 °C digital temperature sensor with I²C interface and ALERT output

Features

- Accuracy: ±0.5 °C maximum from –10 °C to 85 °C
- Accuracy: ±1.0 °C maximum from –40 °C to 125 °C
- 12-bit resolution (0.0625 °C per LSB)
- Supply voltage: 1.7 V to 3.6 V
- Supply current: 3 µA average at one conversion per second
- Shutdown current: 0.5 µA
- I²C interface; four addresses 0x48 to 0x4B selected by the ADD0 pin
- Open-drain ALERT output with programmable limits
- SOT-563 (6-pin) package

General description

The HT410 is a low-power digital temperature sensor for thermal management and compensation. Conversion time is 26 ms typical. An open-drain ALERT output signals when the temperature crosses programmable high and low limits.

Applications

Battery packs, Wearables, Thermal protection, Environmental monitoring.

Ordering information

Order code	Package	Packing	Temperature grade
HT410-SOT563-T	SOT-563 (6-pin)	Tape & reel, 4000	–40 °C to 125 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VDD to GND	–0.3	4.0	V
SDA, SCL, ALERT to GND	–0.3	4.0	V
Storage temperature	–65	150	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Supply voltage VDD	1.7	3.3	3.6	V
I²C clock frequency	—	—	400	kHz
Operating temperature	–40	—	125	°C

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
Temperature accuracy	TA = –10 °C to 85 °C	–0.5	±0.1	0.5	°C
Temperature accuracy	TA = –40 °C to 125 °C	–1.0	±0.2	1.0	°C
Resolution		—	0.0625	—	°C
Conversion time		—	26	35	ms
Average supply current	1 conversion/s, VDD = 3.3 V	—	3	5	µA
Shutdown current		—	0.5	1.5	µA
I²C address (ADD0 = GND)	7-bit		0x48		
I²C address (ADD0 = VDD)	7-bit		0x49		
I²C address (ADD0 = SDA)	7-bit		0x4A		
I²C address (ADD0 = SCL)	7-bit		0x4B		

Pin configuration and functions

Pin	Name	Type	Description
1	SCL	Input	I²C clock.
2	GND	Ground	Ground.
3	ALERT	Output	Open-drain over/under-temperature alert.
4	ADD0	Input	Address select (GND, VDD, SDA or SCL).
5	VDD	Power	1.7 V to 3.6 V supply.
6	SDA	I/O	I²C data (open-drain).

Package and thermal information

Package	Thermal resistance θJA	Dimensions
SOT-563	θJA = 190 °C/W	1.6 × 1.6 mm

Application information

Self-heating is negligible at the specified sampling rate. For best accuracy, thermally couple the package to the surface being measured.

Revision history

Revision	Date	Change
Rev A	2023-11-14	Initial release
Rev B	2024-08-01	Added ±0.5 °C grade